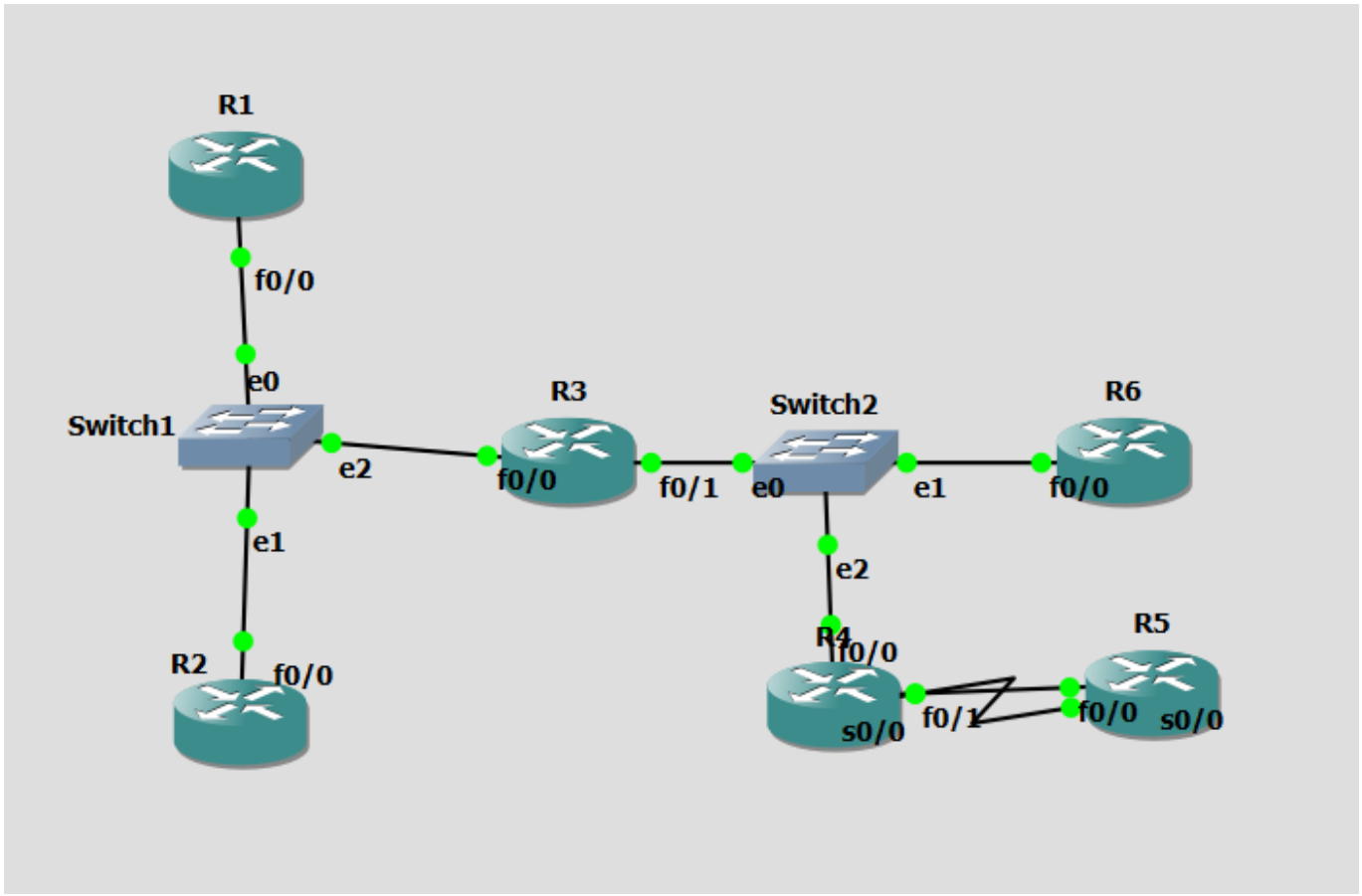
Omar Abdel Motalb | ID: 900232118

Network Topology



Router Link-State Database

```
R1#show ip ospf database router 1.1.1.1

            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

  LS age: 464
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 1.1.1.1
  Advertising Router: 1.1.1.1
  LS Seq Number: 80000004
  Checksum: 0xABA7
  Length: 36
  Number of Links: 1

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.168.123.1
     (Link Data) Router Interface address: 192.168.123.1
      Number of TOS metrics: 0
       TOS 0 Metrics: 10


R1#
```

Router Link-State Database

```
R2#show ip ospf database router 2.2.2.2

            OSPF Router with ID (2.2.2.2) (Process ID 1)

                Router Link States (Area 0)

  LS age: 532
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 2.2.2.2
  Advertising Router: 2.2.2.2
  LS Seq Number: 80000003
  Checksum: 0x6FDB
  Length: 36
  Number of Links: 1

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.168.123.1
     (Link Data) Router Interface address: 192.168.123.2
      Number of TOS metrics: 0
       TOS 0 Metrics: 10


R2#
```

Router Link-State Database

```
R3#show ip ospf database router 3.3.3.3

            OSPF Router with ID (3.3.3.3) (Process ID 1)

                Router Link States (Area 0)

  LS age: 524
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 3.3.3.3
  Advertising Router: 3.3.3.3
  LS Seq Number: 80000002
  Checksum: 0x360C
  Length: 36
  Area Border Router
  Number of Links: 1

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.168.123.1
     (Link Data) Router Interface address: 192.168.123.3
      Number of TOS metrics: 0
       TOS 0 Metrics: 10



                Router Link States (Area 1)

  LS age: 533
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 3.3.3.3
  Advertising Router: 3.3.3.3
  LS Seq Number: 80000003
  Checksum: 0x8359
  Length: 36
  Area Border Router
  Number of Links: 1

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.168.45.2
     (Link Data) Router Interface address: 192.168.45.3
      Number of TOS metrics: 0
       TOS 0 Metrics: 10
```

Router Link-State Database

```
R4#show ip ospf database router 4.4.4.4

            OSPF Router with ID (4.4.4.4) (Process ID 1)

                Router Link States (Area 1)

  LS age: 553
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 4.4.4.4
  Advertising Router: 4.4.4.4
  LS Seq Number: 80000007
  Checksum: 0xAAC0
  Length: 72
  Number of Links: 4

    Link connected to: another Router (point-to-point)
     (Link ID) Neighboring Router ID: 5.5.5.5
     (Link Data) Router Interface address: 192.168.65.1
      Number of TOS metrics: 0
       TOS 0 Metrics: 64

    Link connected to: a Stub Network
     (Link ID) Network/subnet number: 192.168.65.0
 --More--
```

Router Link-State Database

```
R5#show ip ospf database router 5.5.5.5

            OSPF Router with ID (5.5.5.5) (Process ID 1)

                Router Link States (Area 1)

  LS age: 1757
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 5.5.5.5
  Advertising Router: 5.5.5.5
  LS Seq Number: 80000004
  Checksum: 0x1C96
  Length: 60
  Number of Links: 3

    Link connected to: another Router (point-to-point)
     (Link ID) Neighboring Router ID: 4.4.4.4
     (Link Data) Router Interface address: 192.168.65.2
      Number of TOS metrics: 0
       TOS 0 Metrics: 64

    Link connected to: a Stub Network
     (Link ID) Network/subnet number: 192.168.65.0
 --More--
```

Router Link-State Database

```
R6#show ip ospf database router 6.6.6.6

            OSPF Router with ID (6.6.6.6) (Process ID 1)

                Router Link States (Area 1)

  LS age: 576
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 6.6.6.6
  Advertising Router: 6.6.6.6
  LS Seq Number: 80000005
  Checksum: 0x7B4A
  Length: 36
  Number of Links: 1

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.168.45.2
     (Link Data) Router Interface address: 192.168.45.1
      Number of TOS metrics: 0
       TOS 0 Metrics: 10


R6#
```

Neighbors - Default Priority

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DROTHER    00:00:35    192.168.123.2   FastEthernet0/0
3.3.3.3           1   FULL/BDR        00:00:32    192.168.123.3   FastEthernet0/0
R1#
R1#
```

Neighbors - Default Priority

```
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:31    192.168.123.1   FastEthernet0/0
3.3.3.3           1   FULL/BDR        00:00:37    192.168.123.3   FastEthernet0/0
R2#
```

Neighbors - Default Priority

```
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:37    192.168.123.1   FastEthernet0/0
2.2.2.2           1   FULL/DROTHER    00:00:36    192.168.123.2   FastEthernet0/0
4.4.4.4           1   FULL/DR         00:00:36    192.168.45.2    FastEthernet0/1
6.6.6.6           1   FULL/BDR        00:00:37    192.168.45.1    FastEthernet0/1
```

Neighbors - After Priority Change

```
R1(config-if)#end
R1#show ip ospf neighbor
*Mar  1 00:34:11.147: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           5   FULL/BDR        00:00:31    192.168.123.2   FastEthernet0/
0
3.3.3.3           6   FULL/DROTHER    00:00:36    192.168.123.3   FastEthernet0/
0
R1#
```

Neighbors - After Priority Change

```
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           4   FULL/DR         00:00:30    192.168.123.1   FastEthernet0/
0
3.3.3.3           6   FULL/DROTHER    00:00:31    192.168.123.3   FastEthernet0/
0
R2#
```

Neighbors - After Priority Change

```
R3(config-if)#ip ospf priority 6
R3(config-if)#exit
R3(config)#int f0/1
R3(config-if)#ip ospf priority 6
R3(config-if)#end
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           4   FULL/DR         00:00:30    192.168.123.1   FastEthernet0/
0
2.2.2.2           5   FULL/BDR        00:00:37    192.168.123.2   FastEthernet0/
0
4.4.4.4           1   FULL/DR         00:00:32    192.168.45.2    FastEthernet0/
1
6.6.6.6           1   FULL/BDR        00:00:33    192.168.45.1    FastEthernet0/
1
R3#
*Mar  1 00:29:25.727: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

Routing Table

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.123.0/24 is directly connected, FastEthernet0/0
     1.0.0.0/24 is subnetted, 1 subnets
C       1.1.1.0 is directly connected, Loopback0
O IA 192.168.45.0/24 [110/20] via 192.168.123.3, 00:11:22, FastEthernet0/0
O IA 192.168.56.0/24 [110/30] via 192.168.123.3, 00:11:22, FastEthernet0/0
O IA 192.168.65.0/24 [110/84] via 192.168.123.3, 00:11:22, FastEthernet0/0
R1#
```

Routing Table

```
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.123.0/24 is directly connected, FastEthernet0/0
     2.0.0.0/24 is subnetted, 1 subnets
C       2.2.2.0 is directly connected, Loopback2
O IA 192.168.45.0/24 [110/20] via 192.168.123.3, 00:11:52, FastEthernet0/0
O IA 192.168.56.0/24 [110/30] via 192.168.123.3, 00:11:52, FastEthernet0/0
O IA 192.168.65.0/24 [110/84] via 192.168.123.3, 00:11:52, FastEthernet0/0
R2#
```

Routing Table

```
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.123.0/24 is directly connected, FastEthernet0/0
     3.0.0.0/24 is subnetted, 1 subnets
C       3.3.3.0 is directly connected, Loopback3
C    192.168.45.0/24 is directly connected, FastEthernet0/1
O    192.168.56.0/24 [110/20] via 192.168.45.2, 00:12:37, FastEthernet0/1
O    192.168.65.0/24 [110/74] via 192.168.45.2, 00:12:37, FastEthernet0/1
R3#
```

Routing Table

```
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O IA 192.168.123.0/24 [110/20] via 192.168.45.3, 00:12:41, FastEthernet0/0
C    192.168.45.0/24 is directly connected, FastEthernet0/0
     4.0.0.0/24 is subnetted, 1 subnets
C       4.4.4.0 is directly connected, Loopback4
C    192.168.56.0/24 is directly connected, FastEthernet0/1
C    192.168.65.0/24 is directly connected, Serial0/0
R4#
```

Routing Table

```
R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O IA 192.168.123.0/24 [110/30] via 192.168.56.1, 00:13:02, FastEthernet0/0
O    192.168.45.0/24 [110/20] via 192.168.56.1, 00:13:17, FastEthernet0/0
     5.0.0.0/24 is subnetted, 1 subnets
C       5.5.5.0 is directly connected, Loopback5
C    192.168.56.0/24 is directly connected, FastEthernet0/0
C    192.168.65.0/24 is directly connected, Serial0/0
R5#
```

Routing Table

```
R6#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O IA 192.168.123.0/24 [110/20] via 192.168.45.3, 00:13:20, FastEthernet0/0
C    192.168.45.0/24 is directly connected, FastEthernet0/0
     6.0.0.0/24 is subnetted, 1 subnets
C       6.6.6.0 is directly connected, Loopback6
O    192.168.56.0/24 [110/20] via 192.168.45.2, 00:13:35, FastEthernet0/0
O    192.168.65.0/24 [110/74] via 192.168.45.2, 00:13:35, FastEthernet0/0
R6#
```